

AHMED ALHARTHY

Electrical & Computer Engineer

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Portfolio: ahmed-018.github.io | GitHub: github.com/Ahmed-018

SUMMARY

Electrical & Computer Engineering student at King Abdulaziz University (GPA 4.95/5.00) completing a co-op at Saudi Aramco IT Infrastructure. Hands-on with robotics (ROS2), embedded systems, and programming.

EDUCATION

B.Sc. Electrical Engineering (Computer Engineering)

Expected Dec 2026

King Abdulaziz University, Jeddah | GPA 4.95 / 5.00 | Certificate of Excellence

EXPERIENCE

Co-op Trainee, Saudi Aramco - IT Infrastructure Department

Jun 2026 - Present

- Decommissioning legacy telecom equipment at upgraded sites (6 of 30 completed so far).
- Testing communication circuits end to end with a TIMS analyzer to locate faults.
- Maintaining site power systems: 48 VDC rectifiers, battery banks, inverters, and generators.
- Working with the DWDM / OTN optical backbone and fiber network links.

PROJECTS

Automated Solar-Powered Vehicle | Senior Project, Grade A+ | ROS2, Nav2, Python, YOLO, LiDAR

2026

Self-driving, solar-charged vehicle for the KAU campus (team of 3). My part: the navigation and safety software.

- Built the ROS2 / Nav2 navigation: reached the selected destination in 5/5 trials within 1 m.
- Built YOLO human detection that stops the vehicle: detected people in 5/5 trials at up to 11.7 m.
- Tested the system in Gazebo simulation before running it on the real vehicle.

Stock Market Simulator | Advanced Programming | Java, OOP, GUI

2024

Simulated trader bots (RSI, moving average, momentum) and compared their performance.

Ball and Beam Control System | Automatic Control | MATLAB, Simulink

2025

Designed PD and lead controllers that settle in under 3 s with less than 5% overshoot.

8-Bit CPU Design | Digital Design | Verilog, ModelSim

Designed and simulated an 8-bit CPU (datapath, control unit, ALU).

Photos, results and code for each project: ahmed-018.github.io

SKILLS

Programming: Python, C, C++, Java, Verilog, MATLAB

Robotics & AI: ROS2, Nav2, Gazebo, YOLO, OpenCV, Simulink

Hardware & Lab: Arduino, Raspberry Pi, ESP32, FPGA, Oscilloscope, Multimeter, Circuit Simulation

Networking: TCP/IP, IPv4 / IPv6 Addressing, Routing & Switching Fundamentals, Cisco Packet Tracer

Design & Tools: SolidWorks (3D CAD Modeling), AutoCAD, Engineering Drawing & Sketching, 3D Printing, Git, Linux

Office: Microsoft Word, Excel, PowerPoint, Canva

Languages: Arabic (Native), English (Fluent)

CERTIFICATIONS

IELTS Academic - Band 7.5 (C1)

Aug 2026

SCE Fundamentals of Engineering (FE) - Electrical & Computer, in preparation

Feb 2027

VOLUNTEERING

Technical Team, Protothon Hackathon - KAU Makers Club (250 hours)

2025 - 2026

Marketing Team, RoboCup Competition - KAU ECE Club

2024 - 2025

Projects Management Team - KAU ECE Club

2023 - 2025